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Advances in Space Research 60 (2017) 636–651

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

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Effect of grain size distribution on stress-strain behavior of lunar soil simulants

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Received 24 February 2017; accepted 29 April 2017
Available online 11 May 2017

Abstract

Geotechnical behavior of the lunar soils is important for engineering analyses regarding various aspects of the future extraterrestrial settlement plans including lunar exploration and construction. Many lunar soil simulants had been produced so far, in order to resemble lunar soils and conduct such analyses. The goal of this study is to investigate how and to what extent the variations in the grain size distribution of different lunar soil simulants affect their shear strength and volume change behaviors, both of which are quite important for constitutive modeling and geotechnical design. Static simple shear tests were conducted on four lunar soil simulants that were reproduced in terms of original gradation characteristics. The results indicate that various gradational parameters, such as mean grain size, coefficient of uniformity and fines content influence the shear strength, the amount of volumetric dilatancy, and the rate of dilatancy of simulant specimens in different levels when they were compared at the same density or void ratio. The possible reasons behind such different levels of influence were also discussed by focusing on the initial fabric of specimens achieved before shearing and the interaction between silt and sand matrices in the simulants.
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Keywords: Lunar soil simulants; Simple shear test; Shear strength; Internal friction angle; Dilatancy; Soil fabric

1. Introduction

It is not difficult to imagine that developed countries will establish permanent lunar bases in the following few decades in order to explore deeper space. Lunar bases would also be a good trial to extend towards relatively distant planets such as Mars. In fact, countries such as the United States (Benaroya and Bernold, 2008) and Japan (Matsumoto et al., 2006) are already planning to return or access to the moon in order to construct bases. Recently, on December 2013, China successfully landed a lunar rover on the moon as a part of their lunar exploration program (CNSA, 2014).

Likewise on earth, virtually every structure that is necessary for the sustainability of a land base needs to interact with the local regolith or the soil on that particular planet. For lunar exploration, such an interaction is not only important for the design of lunar structures such as laboratories, telescopes, storage and accommodation facilities, power plants, transportation network etc. at the bases, but also important for soft landing on the moon, terramechanics of the lunar rovers, site preparation and radiation shielding. All of the mentioned examples imply the significance of understanding the geotechnical properties of lunar soils for the sustainability of lunar bases and the lunar exploration.

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http://dx.doi.org/10.1016/j.asr.2017.04.029
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According to Arslan et al. (2010a) US Apollo missions brought approximately 115 kg of lunar soil, while Carrier et al. (1973) and Martel (2009) mentioned an approximate amount of 382 kg of lunar rock and soil brought by several missions. In fact, China is planning to bring about 2 kg of lunar regolith to the earth in 2017 by Chang'e 5 mission (CNSA, 2014). Nevertheless, such amounts mentioned are clearly insufficient for conducting extensive experimental programs on various engineering aspects. Therefore, several countries have produced various lunar soil simulants for their space research programs.

Carrier (2003) compiled the grain size distribution curves of over 300 lunar soil samples taken from the vicinity of seven landing sites and mentioned that the grain size distribution of the lunar soil is quite consistent. Accordingly, average mean grain size (D_{50}) of the lunar soil is 0.072 mm, average coefficient of uniformity (CU) is 16 and average coefficient of curvature (CC) is 1.2. Based on the Unified Soil Classification System (USCS) used for terrestrial soils, lunar soil can be classified as silty sand (SM) to sandy silt (ML).

Several researchers have carried out shear strength tests and other experiments in order to characterize the geotechnical properties of the different lunar soil simulants that were intended to mimic the lunar soil. Bernold (1994) conducted direct shear and compaction tests on University of Maryland Lunar Simulant at three different fines contents (i.e. FC = 10%, 30% and 50%). Accordingly, shear strength and compaction efficiency of the simulant have increased with decreasing fines content. Johnson and King (2010) performed direct shear tests on JPL-1 lunar simulant and concluded that a small bucket-wheel excavator would be feasible candidate for lunar soil works (e.g. berms, roads, trenches etc.).

Arslan et al. (2010a) run triaxial compression and direct tension tests on JSC-1A lunar simulant. The observed values of the peak internal friction angle (ϕ_{peak}) were between 42.9° and 48.8° depending on the initial density of the specimens and all specimens were dilatant. Zeng et al. (2010a) also conducted triaxial compression, compaction and 1-D consolidation tests on JSC-1A lunar simulant. They observed ϕ_{peak} values between 41.9° and 56.7° depending on the initial density of the specimens. Iai and Luna (2011) conducted direct shear tests on JSC-1A lunar simulant. The peak internal friction angles were calculated between 34° and 68° depending on the initial density of specimens. Iai and Luna (2011) mentioned that ϕ_{peak} values obtained from direct shear tests were smaller than the values obtained from triaxial tests for loose specimens, whereas vice versa for the dense specimens. In the same study, a relationship is also proposed to relate the internal friction angle of JSC-1A from direct shear tests to the one from triaxial compression tests. All specimens of Iai and Luna (2011), except the very loose one, have shown dilatant behavior during shearing.

Jiang et al. (2012) performed direct shear and 1-D consolidation tests on TJ-1 lunar simulant. Unlike the previous researchers, Jiang et al. (2012) picked a single initial density for their specimens (i.e. $\rho = 1.36$ g/cm^3, $e = 1.0$) and obtained a ϕ_{peak} value of 47.6° for TJ-1. Several other lunar soil simulants such as MLS-1 (Perkins and Madson, 1996), FJS-1 (Kanamori et al., 1998), JSC-1 (Klosky et al., 2000), CAS-1 (Yongchun et al., 2009), NAO-1 (Li et al., 2009), GRC-1 (Oravec et al., 2010), NT-LHT-2M (Zeng et al., 2010b) were also investigated in order to characterize their geotechnical behavior.

Even though all of the mentioned simulants were intended to be produced with high fidelity considering the lunar soil, it is inevitable to have differences among different lunar soil simulants as they are artificially produced. These differences could be in different aspects such as mineralogical composition, toxicity, abrasiveness, geotechnical properties etc. This study focuses on the geotechnical aspect only, as "high fidelity" is a relative concept for different science and engineering disciplines. Yet, geotechnical engineering is especially important for the development and the sustainability of lunar bases (Monkul, 2013a).

One of the important factors that influence the geotechnical behavior of soils is the grain size distribution. It is known to influence the stress-strain behavior of terrestrial silty sands (Monkul, 2013b). In general, with all other influencing factors (e.g. relative density, grain shape, initial stress conditions etc.) being the same, a well graded terrestrial sand (SW) is expected to be stronger than the poorly graded one (SP) (Holtz et al., 2011). On the other hand, Monkul et al. (2016) have recently shown that undrained shear strength of laboratory prepared terrestrial silty sands could significantly decrease with increasing coefficient of uniformity (i.e. as they technically become more well graded). In most of the terrestrial soils pore water could play an important role on the stress-strain behavior depending on the drainage conditions, unlike lunar soils where no water in between the pore space of the soil grains is available. However, similar to the terrestrial soils, there could be variations in grain size distribution of different lunar soil simulants. This paper gives a comparison of the grain size distributions of several lunar soil simulants, but more importantly investigates whether differences in the grain size distributions influence geotechnical behavior such as the stress-strain response and the peak internal friction angle of lunar soil simulants.

2. Grain size distributions of different lunar soil simulants in literature and soils tested in this study

Grain size distribution shows the range and values of the size of individual particles in a soil matrix in terms of percent by dry weight. It is an important characterization tool for soils as it influences many geotechnical properties including, but not limited to shear strength, compressibility, compaction, and permeability in different levels. Grain size distribution of the soils is conventionally determined either by sieve analysis and/or hydrometer tests (ASTM D422, 2005), though other methods are also emerging such

as techniques based on laser diffraction (Li et al., 2009; Yang and Wei, 2012), and X-ray micro-computed tomography (Fonseca et al., 2012).

As mentioned before, several lunar soil simulants were produced and tested in literature in order to mimic the engineering behavior of the lunar soil. Fig. 1 shows the grain size distribution of different lunar soil simulants reported in literature. Even though the gradation curves compiled in Fig. 1 involve the vast majority of the lunar soil simulants produced in last twenty years, it should be noted that there could be some missing ones. Fig. 1 also shows the upper and the lower boundaries for the grain size distribution of the lunar soils brought through Apollo 11, 12, 14 and 15 missions as reported by Carrier et al. (1973). As seen in Fig. 1, the gradation curves of the most simulants fall within this zone.

Table 1 gives various gradational parameters such as fines content (FC), mean grain size (D_{50}), coefficient of uniformity (CU) and Unified Soil Classification System (USCS) group symbol (ASTM D2487, 2005) of the lunar soil simulants shown in Fig. 1. FC refers to the dry weight percent of the particles that are smaller than 0.075 mm in diameter, and shown to be influential on the engineering response of the terrestrial soils, where pore water plays an important role as well (Lade and Yamamuro, 1997; Monkul, 2012). The coefficient of uniformity ($CU = D_{60}/D_{10}$) is one of the basic gradation parameters which gives a general indication of how wide is the range of particle sizes on a grain size distribution curve. For example, as CU decreases the gradation curve of a lunar soil simulant in Fig. 1 becomes steeper. Recent research has shown that there is a strong correlation between the undrianed shear strength of terrestrial silty sands and CU (Monkul et al., 2016). Fines content, CU and some other parameters are used together in order to classify the coarse grained soils (mainly sandy and gravelly soils) for engineering purposes (ASTM D2487, 2005). Table 1 shows that, most of the lunar soil simulants presented in Fig. 1 are silty sands (SM), while few of them are sandy silts (ML), and only one of them is poorly graded sand (SP).

In order to investigate whether the grain size distribution influences the stress-strain response of the lunar soil simulants, two pairs of simulants were selected based on their gradation characteristics analysed in Fig. 1 and Table 1. The first pair is GRC-1 and TJ-1: as seen in Table 1 and Fig. 2a, both simulants have almost identical mean grain size (D_{50}). However, CU of TJ-1 is considerably greater than CU of GRC-1 (about five times). Hence, TJ-1 is relatively well graded compared to GRC-1. Another difference between the two simulants is the amount of fines content: GRC-1 has a FC of 5% (poorly graded sand), while TJ-1 has a FC of 23% (silty sand). The second pair is BP-1 and QH-E, which have almost the same CU (≈12) but different D_{50} and FC (Table 1, Fig. 2b). It should also be noted that even though the difference in the amount of FC between the simulants in Fig. 2b is only 12%, BP-1 is classified as a silty sand (SM), while QH-E is classified as a sandy silt (ML) according to USCS.

Grain size distribution characteristics of the selected two pairs of lunar soil simulants were replicated by mixing a terrestrial sand and a silt. A clean sand was obtained from a sand quarry in Şile region of Istanbul and its grains were separated in to different sizes by sieving. A natural silt was obtained from Izmir, Turkey. Then, targeted simulants were reconstituted by mixing the separated sand grains with the silt at appropriate proportions. The possible effects of several factors including mineralogy, grain shape etc. on the resulting stress-strain behaviors of simulants

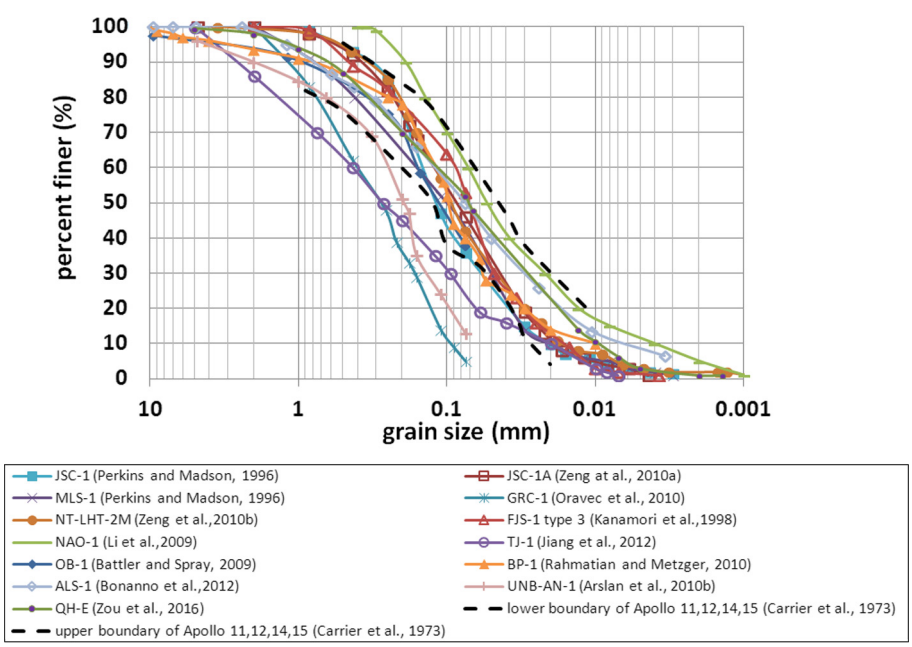

Fig. 1. Grain size distribution curves of different lunar soil simulants reported in literature. (See above-mentioned references for further information.)

Table 1
Some gradational parameters of the lunar soil simulants shown in Fig. 1.

Simulant name	Country of origin	Fines content (%)	D_{50} (mm)	CU (=D_{60}/D_{10})	USCS symbol
JSC-1	USA	36	0.12	7.5	SM
JSC-1A	USA	47	0.086	6.3	SM
MLS-1	USA	43	0.1	7.9	SM
GRC-1	USA	5	0.28	4.2	SP
NT-LHT-2M	USA	42	0.093	6.7	SM
FJS-1	Japan	53	0.071	5.5	ML
NAO-1	China	62	0.054	18	ML
TJ-1	China	23	0.26	21.5	SM
OB-1	Canada	38	0.111	?	SM
BP-1	USA	40	0.098	11.8	SM
ALS-1	Australia	50	0.075	18.5	ML
UNB-AN-1	USA	13	0.19	?	SM
QH-E	China	52	0.071	12.1	ML

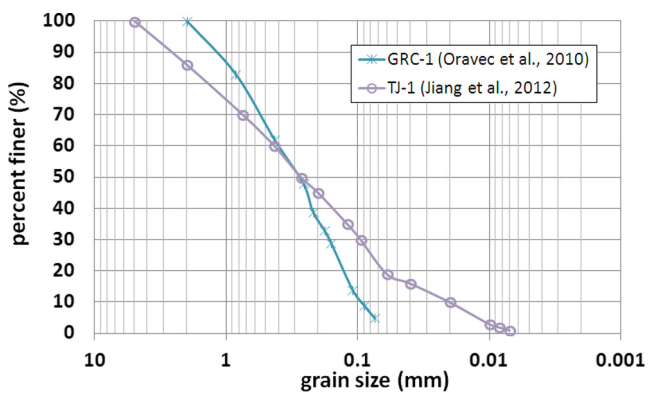

Fig. 2a. Comparison of the grain size distribution curves of the first pair of lunar soil simulants: GRC-1 vs. TJ-1.

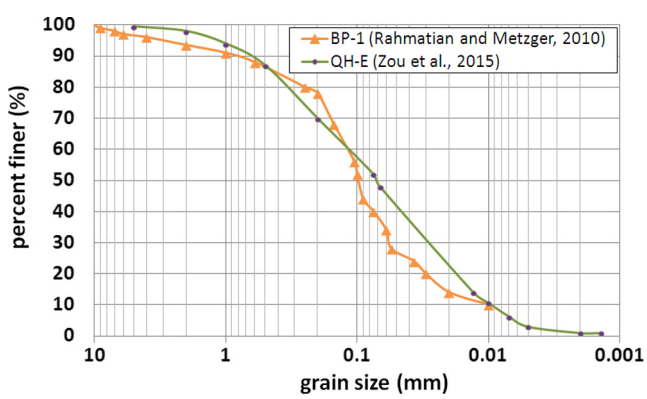

Fig. 2b. Comparison of the grain size distribution curves of the second pair of lunar soil simulants: BP-1 vs. QH-E.

were assumed to be isolated since the same base sand and silt were used for both pairs. The reasons for such a strategy and possible limitations including the bulk chemistry of the proxy simulants are discussed in detail at a separate section (Consideration of limitations, practical implications of findings for planetary exploration and future research) later in the paper.

The reconstituted soils were named as Y-GRC-1, Y-TJ-1, Y-BP-1, Y-QH-E where Y denotes that they were prepared at the soil mechanics laboratory of Yeditepe University. Fig. 3a (Y-GRC-1 vs. Y-TJ-1) and Fig. 3b (Y-BP-1, Y-QH-E) shows the grain size distribution curves of the reconstituted pairs. When Figs. 3a and 3b is compared with Figs. 2a and 2b, it is seen that the grain size distribution curves of the reconstituted pairs match quite well with the original simulant pairs. Table 2 presents a comparison of the gradational parameters between the original lunar soil simulants and the reconstituted ones. As observed in Table 2, values of the gradational parameters of the reconstituted proxy simulants were quite close to the original ones. It should be reminded that it was neither possible, nor necessary to obtain exactly the same grain size distribution considering that silt gradation in the mixtures cannot be adjusted. Nevertheless, the first pair (Y-GRC-1 and Y-TJ-1) that would be tested had almost identical mean grain size ($D_{50} \approx 0.28$ mm) but considerably different CU as targeted (i.e. CU of Y-TJ-1 is six times that of Y-GRC-1). Also Y-GRC-1 has a FC of 5% (poorly graded sand), while Y-TJ-1 has a FC of 23% (silty sand). The second pair that would be tested (Y-BP-1 and Y-QH-E) had identical CU (≈13.5) but different D_{50} and FC (Table 2). Also Y-BP-1 is classified as silty sand (SM), while Y-QH-E is classified as sandy silt (ML) according to USCS.

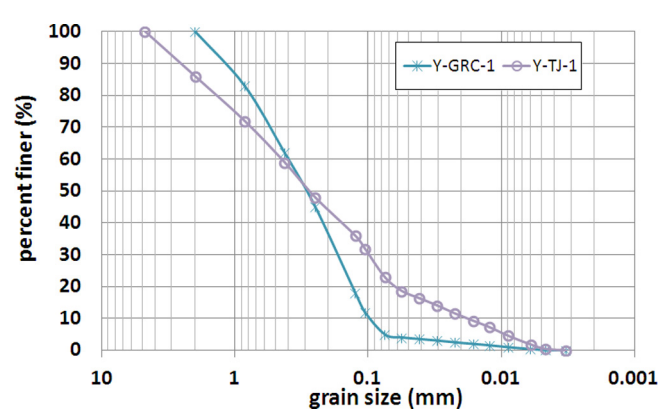

Fig. 3a. Comparison of the grain size distribution curves of the first pair of reconstituted simulants: Y-GRC-1 vs. Y-TJ-1.

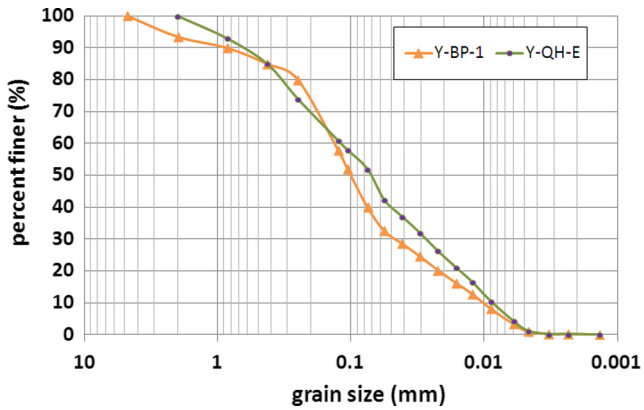

Fig. 3b. Comparison of the grain size distribution curves of the second pair of reconstituted simulants: Y-BP-1 vs. Y-QH-E.

Table 2
Some gradational parameters of the lunar soil simulants shown in Figs. 2a, 2b and 3a, 3b.

Simulant name	FC (%)	D_{50} (mm)	CU (=D_{60}/D_{10})	USCS symbol
GRC-1	5	0.28	4.2	SP
Y-GRC-1	5	0.29	4.1	SP
TJ-1	23	0.26	21.5	SM
Y-TJ-1	23	0.28	25	SM
BP-1	40	0.098	11.8	SM
Y-BP-1	40	0.099	13.5	SM
QH-E	52	0.071	12.1	ML
Y-QH-E	52	0.07	13.6	ML

Specific gravity tests according to ASTM D854 (2005) were performed on both the sand and the silt matrices used in the reconstituted simulants, and it was determined that sand has a specific gravity (G_s) of 2.65, while silt has a G_s of 2.70. These numbers might seem low compared to the values reported in literature for most lunar soil simulants which were around 2.9 to 3.2. However, recently Jiang et al. (2012) added silicon carbide to lunar soil simulant TJ-1 and increased its specific gravity from 2.72 to 2.90 and reported that the change in mechanical behavior and shear strength parameters (e.g. ϕ', c′) of the simulant due to specific gravity increase was negligible. Therefore, the relatively lower G_s values of the soils tested in this study are not expected to influence the resulting mechanical response of the reconstituted proxy simulants.

3. Experimental program

3.1. Static simple shear testing

Direct shear and triaxial compression tests are perhaps the most common high strain element tests used for determination of the shear strength of soils. In this study, another alternative: static simple shear testing was preferred to investigate the stress-strain behavior of the reconstituted lunar soil simulants. The stress conditions of static simple shear testing were shown in Fig. 4; where σ_n corresponds to the normal stress under which the simulants were consolidated and sheared, σ_{lat} corresponds to the lateral stress, and K_o is the coefficient of lateral earth pressure at rest condition. Consolidation in simple shear is anisotropic as shown in Fig. 4a, and can be assumed to represent at rest condition in the field (i.e. $\sigma_{lat} = K_o . \sigma_n$). During the shearing stage, complementary shear stresses on the sides of specimen are absent as shear stresses (τ) are only applied on the top and bottom surfaces (Fig. 4b), yet a fairly homogeneous state of shear stress is applied when compared with

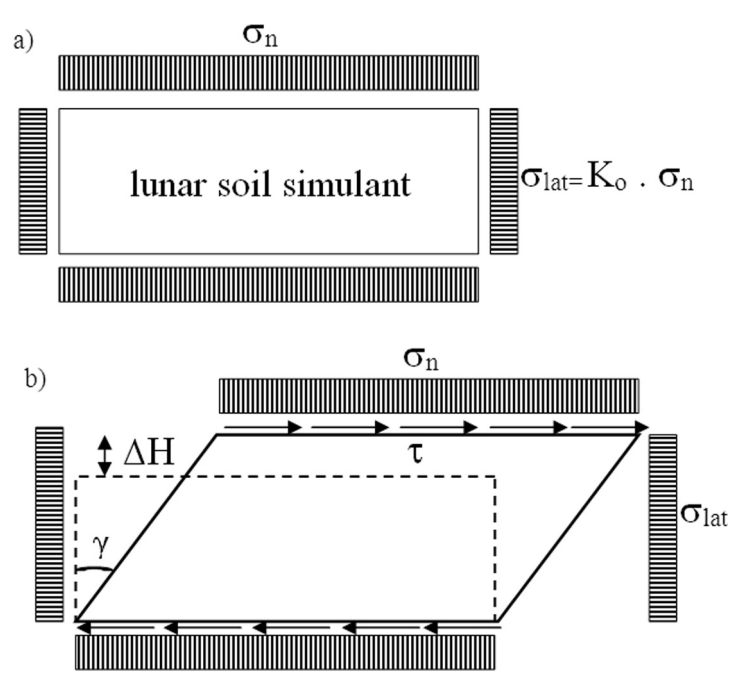

Fig. 4. Stress conditions in a drained static simple shear test; (a) at consolidation stage, (b) at shearing stage for a dilative specimen.

the direct shear test (Holtz et al., 2011). Also specimen preparation is relatively easier compared to triaxial testing, mainly due to the smaller size of simple shear specimens with respect to the typical triaxial test specimens.

In this study, an NGI type static simple shear device at Yeditepe University was used and the tests were done in drained condition (with dry specimens). The typical specimens have a diameter of 64 mm and a height of 20 mm. The lateral confinement was provided by aligning several teflon coated rings around a conventional latex membrane. Because the specimen cross-sectional area is constant during shearing (Fig. 4), volumetric strain (ε_v) is equal to the axial strain (ε_a) in a simple shear test. The amount of shear strain (γ) is also measured as shown in Fig. 4b.

Table 3 shows the type of shear strength tests, consolidation stresses, the specimen densities (ρ) for some of the lunar soil simulants investigated in literature. As seen in Table 3, testing stresses in literature had a wide range changing from as low as 1 kPa to as high as 625 kPa. Assuming that gravity on the moon is about one sixth that of the earth (i.e. $g = 1.62\ m/s^2$), relatively lower normal stresses of 10 kPa, 20 kPa and 40 kPa were selected for the static simple shear tests in this study. These stress values correspond to depths of 3.6 m, 7.3 m and 14.6 m below the lunar surface respectively, when no surface loading was considered. Note that the mentioned values of depths are approximate and calculated based on the assumption that the density of the lunar soil is 1.68 g/cm^3 based on the drill core samples from different Apollo missions (Carrier et al., 1991), however the actual density values at different sites could change with depth and location.

3.2. Specimen preparation

Another important parameter for comparing the stress-strain behavior of lunar soil simulants is the specimen density. As seen in Table 3, most of the studies were performed on at least two bulk densities, which generally correspond to medium dense and dense states for that simulants. In fact, there had been different preferences in literature

Table 3
Shear strength testing conditions for some lunar soil simulants in literature.

Simulant name & reference	Type of strength test	Testing stresses* (kPa)	Avg. ρ of tested simulants (g/cm^3)	Cohesion (kPa)	ϕ_{peak} (deg.)
JSC-1A (Alshibli and Hasan, 2009)	Triaxial compression test	10, 20, 40, 80, 200	1.63	2.0	40
			1.88	5.0	49
JSC-1A (Arslan et al., 2010a)	Triaxial compression test	15, 30, 60	1.7	1.4	42.9
			1.8	1.8	44.5
			1.9	2.4	48.8
JSC-1A (Zeng et al., 2010a)	Triaxial compression test	100, 150, 200	1.66	Not specified	41.9
			1.79		46.5
			1.94		56.7
JSC-1A (Iai and Luna, 2011)	Direct shear test	8, 16, 33, 66	1.59	0.8	34.1
			1.73	0.8	46.2
			1.85	5.6	52.8
			>2.01	37.5	68.1
JPL-1 (Johnson and King, 2010)	Direct shear test	Not specified	1.52	5.38	36
Unv. of Maryland lunar simulant (Bernold, 1994)	Direct shear test	20, 45, 100	1.9 (with 50%FC)	14	39
			1.97 (with 10%FC)	13	41
TJ-1 (Jiang et al., 2012)	Direct shear test	50, 100, 150, 200, 300	1.36	0.86	47.6
MLS-1 (Perkins and Madson, 1996)	Triaxial compression test	10	2.2	0	58[a]
		40		0	55
		625		0	50
JSC-1 (Klosky et al., 2000)	Triaxial compression test	1, 2, 5, 10, 20, 40, 80	1.62	3.9	44.4
			1.72	6.2	49.5
			1.81	14.4	53.6
NAO-1 (Li et al., 2009)	Triaxial compression test	100, 200, 300, 400	1.93	95.3	46.6
GRC-1 (Oravec et al., 2010)	Triaxial compression test	50, 100, 200	1.66[b]	Not specified	33.4
			1.76		38.4
			1.82		44.4
NT-LHT-2M (Zeng et al., 2010b)	Triaxial compression test	100, 150, 200	1.65	0	36
			1.75	0	38.5
			1.87	0	40.7
QH-E (Zou et al., 2016)	Triaxial compression test	3.13, 6.25, 12.5, 25	1.58	4.1	45.1
			1.64	3.1	51.6
			1.71	5.2	54.5

* The stress values correspond to normal stress (σ_n) for direct shear tests, confining stress (σ_c) for triaxial compression tests.
[a] These ϕ_{peak} corresponds to the secant friction angle calculated by Perkins and Madson (1996) under corresponding σ_c.
[b] Tests at several other densities were also performed in the original study.

regarding the appropriate comparison parameter for the stress-strain behavior of terrestrial sandy soils. These preferences also include the parameters of void ratio (e), intergranular void ratio (e_s), relative density (D_r), loosest possible density after deposition (Monkul and Yamamuro, 2011; Monkul et al., 2015). Authors believe that it is difficult to claim that one of the mentioned comparison bases/parameters is superior or more suitable than the others. Instead, all of them are legitimate with proper consideration of the conditions. Nevertheless, in this study specimen density is selected as the comparison basis and all the reconstituted lunar soil simulants were intended to be deposited at the same initial bulk density of 1.68 g/cm^3 (i.e. $\rho = 1.68$ g/cm^3), considering the linear approximation of the bulk density values against depth below the lunar surface obtained by drill core samples from different Apollo missions (Carrier et al., 1991). Note that this value is also within the density range of the vast majority of the tested simulants in literature (Table 3).

The dry funnel deposition technique, explained in Monkul et al. (2015), is used in order to prepare simple shear specimens. A bottle shaped funnel with a curved tip was placed on the porous stone at the base platen, and it was filled with dry soil. Then the funnel was raised gently along the axis of symmetry of the specimen allowing the soil to slowly fill into the space inside the rings with no drop height as schematically shown in Fig. 5. Once the initial deposition stage was completed, top cap and the attached piston (a total mass of 1.012 kg), which would be used on the shearing stage, were gently placed on the specimens as a preseating pressure. Then if necessary, tamping was applied on the specimens to reach the target initial bulk density of 1.68 g/cm^3, which was determined based on the height of the specimen immediately before the consolidation stage. After the deposition process was completed, all specimens were consolidated to one of the mentioned normal stress (σ_n) values (either 10 kPa, 20 kPa, or 40 kPa), and sheared in simple shear condition with a rate of 0.5 mm/min.

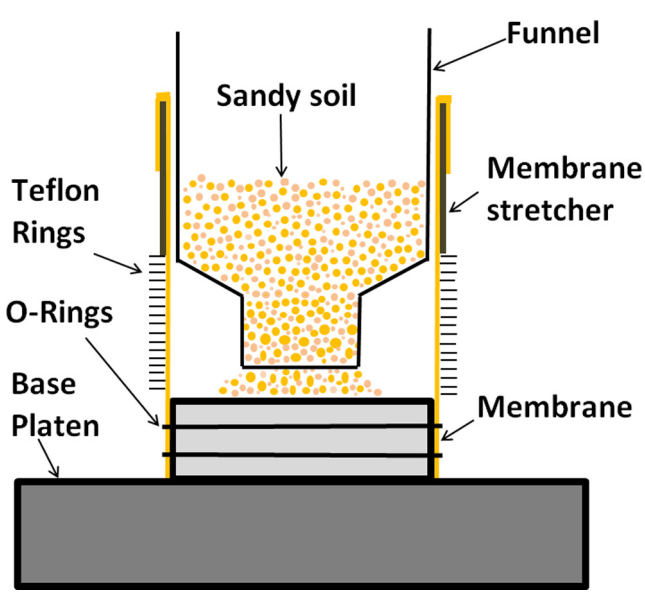

Fig. 5. Specimen preparation with dry funnel deposition.

4. Stress-strain behavior of tested lunar soil simulants

4.1. Drained shear strength characteristics of tested proxy lunar soil simulants

Shear stress-shear strain response of the first pair of the proxy lunar soil simulants are shown in Fig. 6a (for Y-GRC-1) and 6b (for Y-TJ-1). Similarly, the shear stress-shear strain response of the second pair of the proxy lunar soil simulants are shown in Fig. 7a (for Y-BP-1) and 7b (for Y-QH-E). As expected from classical soil mechanics knowledge, shear strength (τ_{max}) of the simulants has increased with increasing normal stress (σ_n). The shear stress- shear strain responses of both simulant pairs were observed to be ductile. In fact, shear stress-shear strain behavior of medium dense to dense sands in drained static simple shear testing were observed to be quite ductile in literature (Kim, 2009), unlike the relatively brittle response of the deviator stress (σ_1–σ_3) or principal stress ratio (σ'_1/σ'_3) with axial strain typically observed for medium dense to

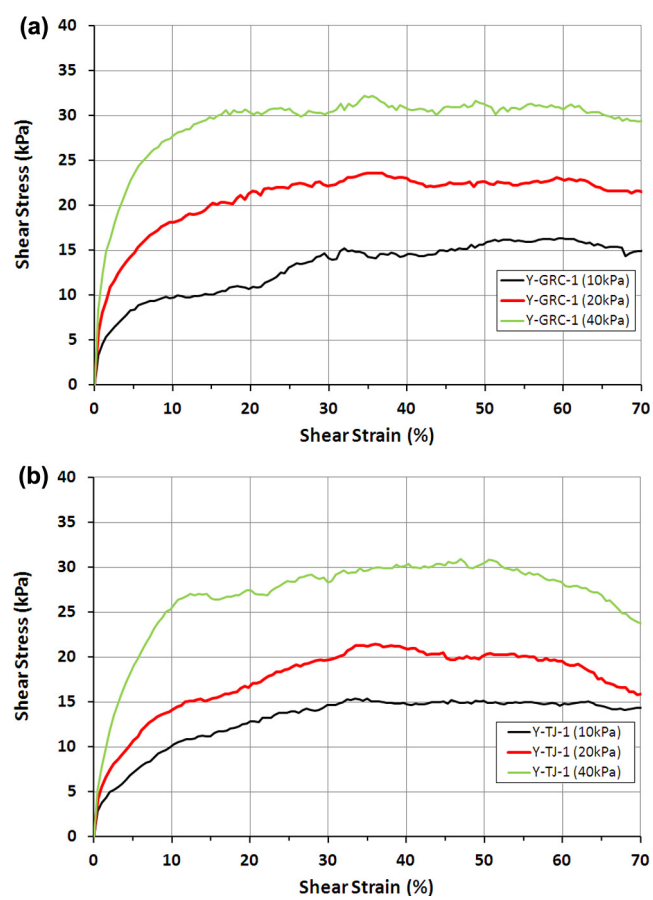

Fig. 6. Shear stress-shear strain responses of (a) Y-GRC-1 lunar soil simulant, (b) Y-TJ-1 lunar soil simulant under different normal stresses.

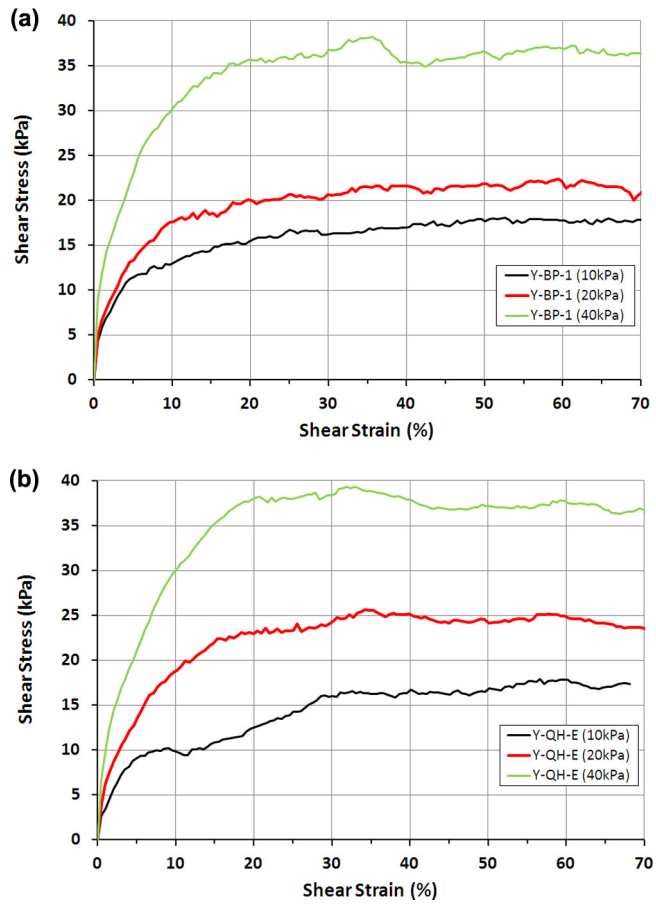

Fig. 7. Shear stress-shear strain responses of (a) Y-BP-1 lunar soil simulant, (b) Y-QH-E lunar soil simulant under different normal stresses.

dense sands in drained triaxial compression tests (Lee and Seed, 1967).

Various failure criteria are available for soils, but Mohr-Coulomb failure criterion given in Eq. (1) is the most popular and simple criterion to express the failure in soils. Where shear strength (τ_f) is expressed as a function of cohesion (c), normal stress (σ_n) and internal friction angle of the soil (ϕ).

$$\tau_f = c + \sigma_n \cdot \tan\phi \quad (1)$$

Lade (1984) mentioned three independent characteristics of failure surfaces for frictional materials as (1) the opening angle of the failure surface, (2) the curvature of the failure surface, (3) the tensile strength of the material. The opening angle is represented by ϕ, and the tensile strength is represented by c in Eq. (1), however the curved nature of the failure surface is not represented in Eq. (1). On the contrary, Eq. (1) assumes that the failure surface is linear. Fig. 8a shows the Mohr-Coulomb failure surface for the first pair of the lunar soil simulants (Y-GRC-1 vs. Y-TJ-1) at the peak state. Interestingly, the failure surfaces of both simulants are parallel to each other having almost the same peak friction angle (i.e. 27.2° vs. 27.1°). Similarly, the cohesion intercepts of the both simulants are reasonably close to each other (i.e. 12.1 kPa vs. 10.1 kPa) as

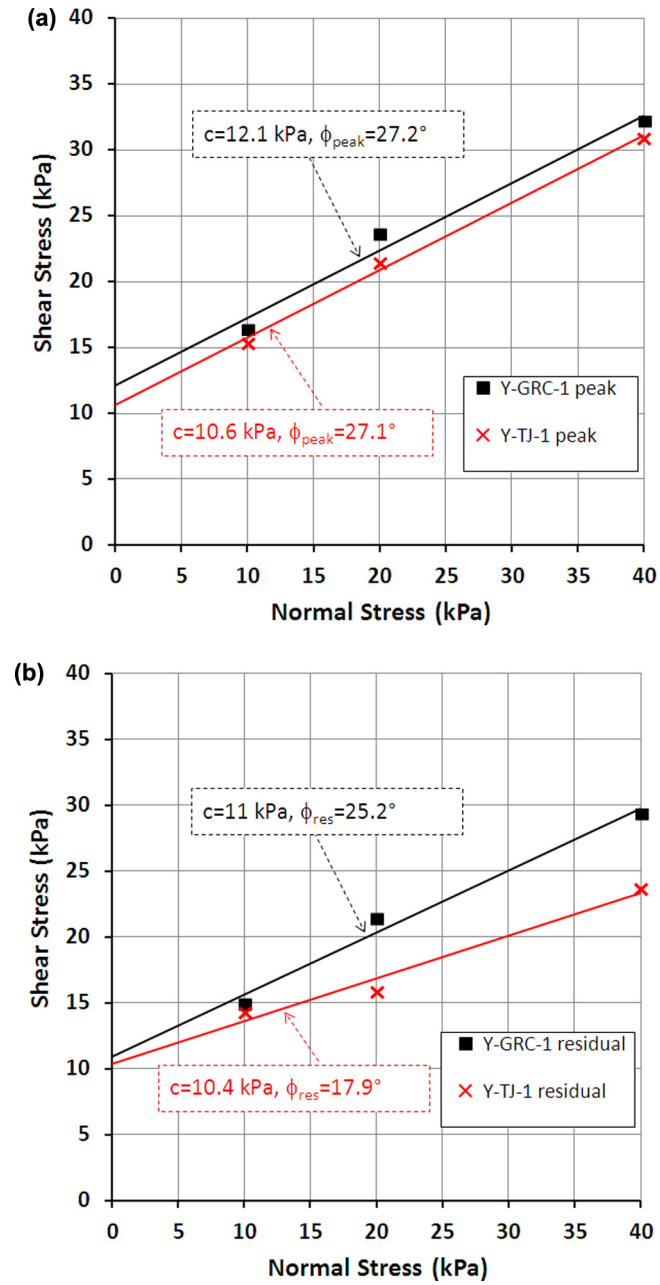

Fig. 8. Mohr-Coulomb envelopes for the first pair of lunar soil simulants (Y-GRC-1 vs. Y-TJ-1) (a) at the peak state, (b) at the residual state (i.e. $\gamma = 70\%$).

shown in Fig. 8a, which makes the failure surface of Y-GRC-1 lie only slightly above that of Y-TJ-1. Alternatively, Mohr-Coulomb relationship for the first pair of the lunar soil simulants at the residual state is shown in Fig. 8b, where residual state is considered at 70% shear strain. One can see that the drop in internal friction angle of Y-GRC-1 from failure to residual state is minor (only 2°), whereas the drop for Y-TJ-1 is much considerable (9.2°).

Fig. 9a shows the Mohr-Coulomb failure surface for the second pair of the lunar soil simulants (Y-BP-1 vs. Y-QH-E) at the peak state. Similar to the first pair, the failure

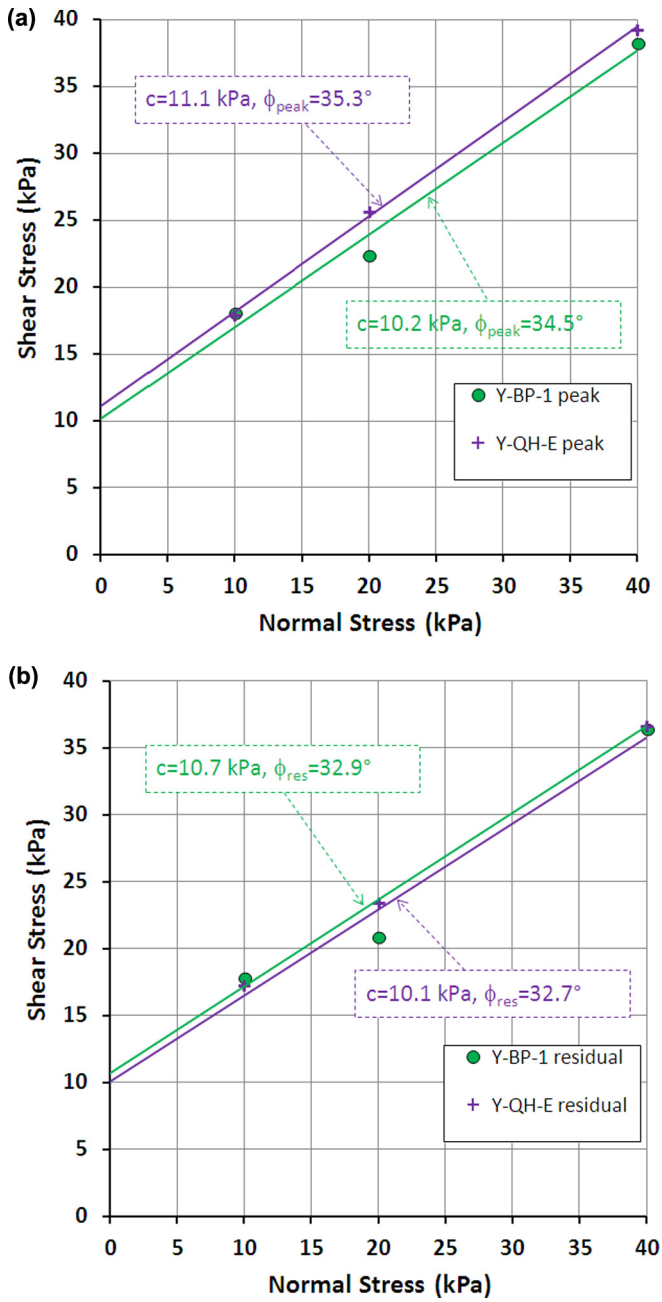

Fig. 9. Mohr-Coulomb envelopes for the second pair of lunar soil simulants (Y-BP-1 vs. Y-QH-E) (a) at the peak state, (b) at the residual state (i.e. $g = 70\%$).

surfaces of both simulants are parallel to each other having almost the same peak friction angle (i.e. 34.5° vs. 35.3°). Also, the cohesion intercepts of the both simulants are reasonably close to each other (i.e. 10.2 kPa vs. 11.1 kPa) as shown in Fig. 9a. At the residual state, Mohr-Coulomb failure surfaces for both simulants almost merged to each other with more or less identical c and ϕ_{res} values as shown in Fig. 9b, where residual state is considered at 70% shear strain.

Considering that cohesionless sand and non-plastic silt mixtures were used in the experimental program, one would initially expect greater peak friction angle values and smaller cohesion values than the ones obtained in Figs. 8 and 9, even though cohesion values of 13 kPa (Bernold, 1994), 14.4 kPa (Klosky et al., 2000), 37.5 kPa (Iai and Luna, 2011) and 95.3 kPa (Li et al., 2009) were reported for various lunar soil simulants in Table 3. Authors think that there are two main reasons behind such relatively high cohesion values. The first one is over-compacting the specimens during the densification process to reach high initial target density values (corresponding to medium dense to dense states), as also mentioned by Iai and Luna (2011). This trend is clearly seen in cohesion column of Table 3, where cohesion values were observed to increase with increasing specimen density for a particular simulant. The second one is the linear approximation of the real curved failure envelope by the Mohr-Coulomb failure criterion given in Eq. (1), as mentioned before. This can be better visualized in Fig. 10, where a schematic representation is given for a normally consolidated cohesionless silty sand (SM) in dry condition for example.

In fact, the definition of cohesion should be reminded in order to make a better interpretation. Mitchell (1993) explained the cohesive strength in two types as true cohesion, and apparent cohesion. The true cohesion occurs usually as a result of chemical bonding between the soil grains, such as due to cementation. This type of cohesion is the one forming tensile strength of the soils, which is the third independent characteristic of a failure surface stated by Lade (1984) as mentioned before. The apparent cohesion on the other hand, is not due to interparticle bonding but due to physical interparticle forces such as capillary stresses or interlocking of grains due to overcompaction for example. Nevertheless, as Mitchell (1993) mentioned, the precise values of either true or apparent cohesion of soils is difficult to determine because of the uncertainty of the intercept when the curved failure envelope is projected backwards to the $\sigma_n = 0$. Therefore, the cohesion and friction angle values obtained in Figs. 8 and 9, should be treated more like as line fitting parameters because of the discussed reasons.

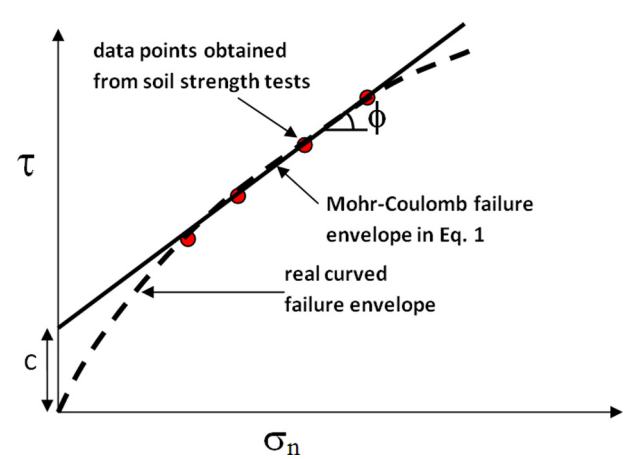

Fig. 10. Schematic representation of a Mohr-Coulomb failure envelope versus the real curved failure envelope for a normally consolidated cohesionless silty sand in dry condition.

Another approach could be assuming the cohesion of simulants as zero, and calculating the secant friction angles with changing confining or normal stress to reflect the curvature of the real failure envelope (Perkins and Madson, 1996; Alshibli and Hasan, 2009). The changes of the peak secant friction angles ($\phi_{sec\text{-}peak}$) with normal stress for the tested simulants in this study are plotted in Fig. 11 with corresponding R^2 values. Accordingly, there is a very smooth relationship of decreasing $\phi_{sec\text{-}peak}$ with increasing σ_n in Fig. 11, which reflects the curvature of the real failure envelope for the simulants such as the one schematically shown in Fig. 10.

When the first pair of simulants (Y-GRC-1 vs. Y-TJ-1) is considered, it could be seen in Fig. 11 that the trend curve for Y-GRC-1 lies slightly above that of the Y-TJ-1 with an approximate difference of 1.5° in terms of friction angle. More specifically, the value of $\phi_{sec\text{-}peak}$ is 56.9° and 58.6° when $\sigma_n = 10$ kPa, and its value is reduced to 37.7° and 38.8° when $\sigma_n = 40$ kPa for Y-TJ-1 and Y-GRC-1 respectively. It should be reminded that Y-GRC-1 and Y-TJ-1 had almost identical mean grain size ($D_{50} \approx 0.28$ mm) but considerably different CU (i.e. CU of Y-TJ-1 is six times that of Y-GRC-1) (see Fig. 3a and Table 2). In other words, even though they have the same D_{50}, Y-TJ-1 is considerably well graded compared to Y-GRC-1. In fact, soils with CU greater than 15 are considered as very well graded (Holtz et al., 2011), and as mentioned before, with other influencing factors being the same, a well graded terrestrial sand (SW) is expected to be stronger than the poorly graded one (SP). However in the case of Y-GRC-1 (SP) versus Y-TJ-1 (SM), it seems that possible influence of coefficient of uniformity to increase the drained strength is counterbalanced with the fines content effect, when the samples are prepared at the same initial density of 1.68 g/cm^3. More specifically, even though Y-TJ-1 is very well graded compared to Y-GRC-1, because of the considerably greater fines content in it (i.e. FC = 23%), its drained strength is even slightly smaller than that of Y-GRC-1 (with FC = 5%) (Fig. 11). The effect of fines on the initial fabric of the simulant specimens would be further discussed in the proceeding section of the paper.

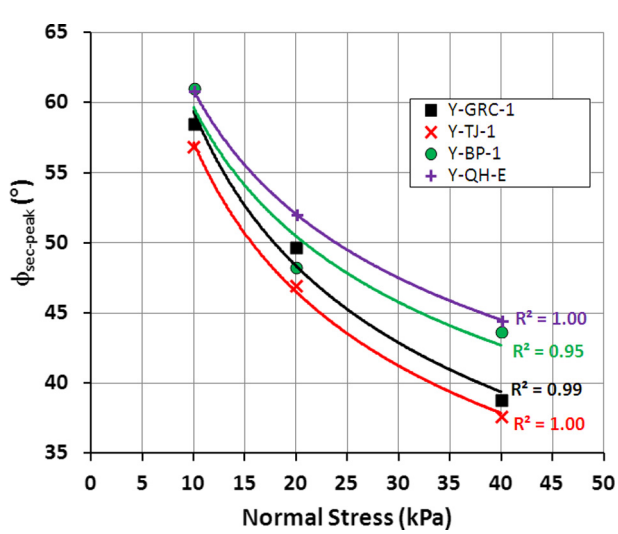

Fig. 11. The change of peak secant friction angles ($\phi_{sec\text{-}peak}$) with normal stress (σ_n) for the tested simulants in this study.

When the second pair of simulants (Y-BP-1 vs. Y-QH-E) is considered, it could be seen in Fig. 11 that this time the trend curve for Y-QH-E lies slightly above that of the Y-BP-1. The value of $\phi_{sec\text{-}peak}$ is reduced from the vicinity of 60° to the vicinity of 44° when the normal stress increased from 10 kPa to 40 kPa. It should be reminded that Y-BP-1 and Y-QH-E had identical coefficient of uniformities (CU ≈13.5) but different D_{50} values and FC (see Fig. 3b and Table 2). In the case of Y-BP-1 (SM) versus Y-QH-E (ML), it seems that the smaller D_{50} and greater fines content of Y-QH-E made it slightly stronger than Y-BP-1 when the samples are prepared at the same initial density.

When two pairs of simulants are compared, one can observe that the second pair of simulants (Y-BP-1 and Y-QH-E), which have considerably greater fines content and considerably smaller D_{50}, are stronger than the first pair of simulants (Y-GRC-1 vs. Y-TJ-1) when compared at the same initial density (Fig. 11).

4.2. Volume change characteristics of tested proxy lunar soil simulants

The change of volumetric strain (ε_v) with shear strain (γ) for the first pair of proxy lunar soil simulants are plotted in Figs. 12a and b under normal stresses of 10 kPa and 40 kPa respectively. It was observed in both figures that at the initial stages of simple shearing all specimens had shown volumetric contraction then started to dilate. Also in parallel to the classical soil mechanics knowledge, as the normal stress applied on the specimens was increased, the amount of volumetric contraction at the initial stages of shearing increased, meanwhile the following volumetric dilation has decreased. In other words, ε_v vs. γ response for a particular lunar soil simulant has shifted down in Fig. 12b when compared to Fig. 12a because of increasing σ_n.

Fig. 12a and b also clearly shows that Y-GRC-1 specimens were considerably more dilative than the Y-TJ-1 specimens regardless of the normal stress. Table 4 gives the values of several density index parameters including initial and consolidated densities and void ratios of the simulant specimens shown in Fig. 12. Void ratio (e) is a very classical and commonly used parameter in soil mechanics, which is the ratio of volume of the voids (V_v) to the volume of solids (V_s) in a soil body (Eq. (2)). It could be seen in Table 4 that the consolidated void ratio (e_c) values of Y-GRC-1 specimens were greater than the ones for Y-TJ-1 specimens. This means that had the soils in Fig. 12 been tested at exactly the same void ratio (e.g. e_c of the Y-TJ-1), Y-GRC-1 specimens would even be more dilative than shown in Fig. 12 (i.e. the volumetric strain curves of Y-GRC-1 specimens would shift upwards). This is due to

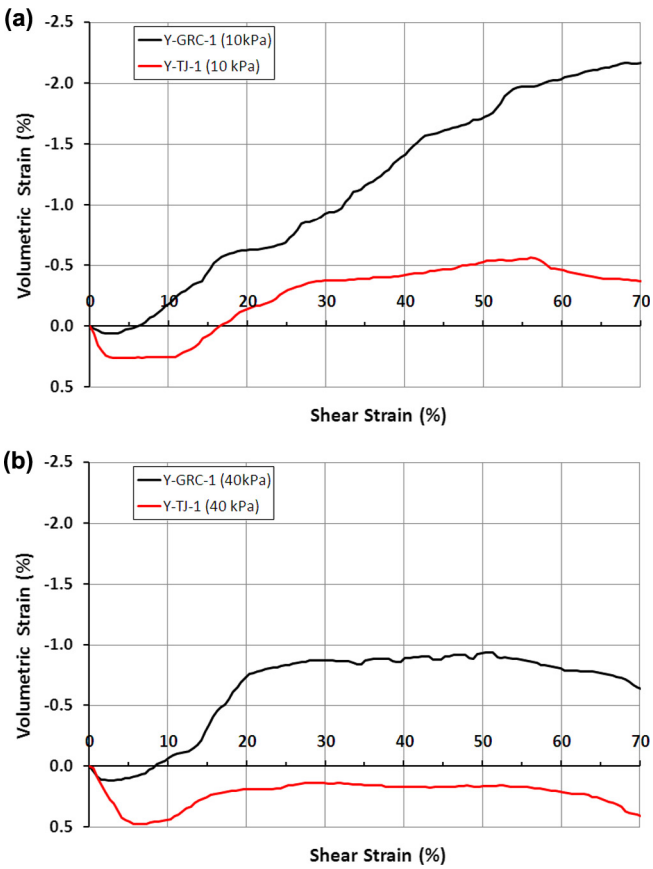

Fig. 12. Volumetric strain-shear strain responses of the first pair of lunar soil simulants: Y-GRC-1 and Y-TJ-1 (a) under $\sigma_n = 10$ kPa, (b) under $\sigma_n = 40$ kPa.

the well known fact that a particular soil becomes more dilatant (or less contractive) as its void ratio decreases. Consequently, one can clearly state that Y-GRC-1 specimens are considerably more dilative than Y-TJ-1 specimens when compared either at the same density or at the same void ratio.

$$e = \frac{V_v}{V_s} \tag{2}$$

This is an interesting observation in a sense that one might possibly expect the relatively very well graded Y-TJ-1 specimens (with CU = 25) to be more dilative than the Y-GRC-1 specimens (with CU = 4.1). This is because the greater coefficient of uniformity (CU) is typically expected to increase the volumetric dilation via more efficient packing tendency and interlocking effect between the grains. One of the possible reasons why such an explained expectation did not happen could be fines content effect that could influence the initial fabric of the tested lunar soil simulants. According to the previous literature on terrestrial silty sands, silt grains in the simulants are expected to have two main positional tendencies: they would either locate in to the intergranular void space in between the surrounding sand grains, or they would locate between the contact points of sand grains by pushing the sand grains apart (Lade and Yamamuro, 1997; Monkul, 2012). For specimens of silt-sand mixtures such as the ones analyzed in Fig. 12, both of the mentioned positional tendencies exist simultaneously. Therefore, the proportion and balance between these two main tendencies is expected to influence the volume change behavior of simulants during shearing.

The condition of the sand grain matrix within the simulants can be expressed by the parameter of intergranular void ratio. Intergranular void ratio (e_s) is the void ratio of the sand matrix in a soil when the silt grains (or other fines) are assumed as voids (Mitchell, 1993; Pitman et al., 1994; Lade and Yamamuro, 1997), with its basic form of equation is given in Eq. (3a). Monkul and Ozden (2007) suggested Eq. (3b) in order to calculate intergranular void ratio, where G_s, G_f, and G are specific gravities of the sand matrix, fines and the overall soil respectively. Table 4 shows that intergranular void ratio values (after consolidation) for Y-TJ-1 simulants are noticeably greater than the values for Y-GRC-1. In other words, the sand grain matrix in Y-TJ-1 specimens before shearing was in a significantly looser configuration than the sand grain matrix in Y-GRC-1 specimens, even though the latter had greater overall void ratio (Table 4). This observations confirms the two main positional tendencies of the silt grains as explained before (i.e. silt grains located in the intergranular voids of Y-TJ-1 makes its void ratio smaller, meanwhile silt grains located between the contact points of the sand grains in Y-TJ-1 makes its intergranular void ratio greater than Y-GRC-1 specimens). Once the specimens were sheared, the silt grains located in the vicinity of sand grain contacts of

Table 4
Values of several density index parameters for the tested simulants.

Tested simulant	Initial density (g/cm^3)	Initial void ratio, e_o	Consolidated density (g/cm^3)	Consolidated void ratio, e_c	Intergranular void ratio, e_s
Y-GRC-1 (10 kPa)	1.69	0.565	1.71	0.548	0.629
Y-GRC-1 (40 kPa)	1.68	0.579	1.71	0.545	0.626
Y-TJ-1 (10 kPa)	1.71	0.560	1.74	0.533	0.982
Y-TJ-1 (40 kPa)	1.69	0.572	1.74	0.529	0.977
Y-BP-1 (10 kPa)	1.68	0.594	1.72	0.549	1.562
Y-BP-1 (40 kPa)	1.68	0.589	1.72	0.554	1.570
Y-QH-E (10 kPa)	1.68	0.596	1.69	0.582	2.264
Y-QH-E (40 kPa)	1.68	0.595	1.71	0.565	2.229

Y-TJ-1 were pushed in to the intergranular voids and together with the re-configuration of the sand grain matrix, Y-TJ-1 specimens in Fig. 12 show much more contractive (or less dilative) volumetric response as opposed to Y-GRC-1 specimens at the same density.

$$e_s = \frac{V_v + V_{fines}}{V_{sand}} \tag{3a}$$

$$e_s = \frac{e + \frac{G}{G_f} \cdot \frac{FC}{100}}{\frac{G}{G_s} \cdot (1 - \frac{FC}{100})} \tag{3b}$$

The change of volumetric strain (ε_v) with shear strain (γ) for the second pair of the proxy lunar soil simulants are plotted in Fig. 13a and b under normal stresses of 10 kPa and 40 kPa respectively. In general, several similar trends to the ones in Fig. 12 were observed, i.e. at the initial stages of simple shearing all specimens shown volumetric contraction then started to dilate; as the normal stress applied on the specimens was increased, the amount of volumetric contraction at the initial stages of shearing increased, meanwhile the following volumetric dilation has decreased.

Fig. 13a and b also shows that Y-BP-1 specimens were more dilative than Y-QH-E specimens regardless of the normal stress. It should be reminded that this pair of simulants had the same coefficient of uniformity (CU), but different FC and D_{50} (Table 2, Fig. 3b). It seems that when prepared at the same initial density and having the same CU, as the D_{50} of specimens in Fig. 13 decreased (in parallel, as the FC of specimens increased), the simulants became less dilative. The trend of decreasing dilatancy with increasing fines content of both simulant pairs in Figs. 12 and 13 coincides well. However, it should be noted that the initial fabric of specimens for the second simulant pair (Fig. 13) are different than the initial fabric and depositional tendencies explained for the first simulant pair (Fig. 12). For the second simulant pair (Y-BP-1 & Y-QH-E) the amount of fines content is so high that virtually all the sand grains should have been separated and float within the silt grain matrix with no or very little sand to sand contact. This can be verified by investigating the density index parameters given in Table 4. Unlike the first pair of the proxy simulants (Y-GRC-1 & Y-TJ-1), where void ratio decreases and intergranular void ratio increases simultaneously with increasing FC, for the second pair (Y-BP-1 & Y-QH-E) values of both e_c and e_s increases markedly with increasing FC. In fact, intergranular void ratio has no practical meaning at such high fines contents (see the unreasonably high values of e_s for Y-BP-1 and Y-QH-E), where sand grains float within the silt grains.

The rate of volumetric dilation of soil could be expressed by dilatancy angle (ψ) which corresponds to the slope of the volumetric strain versus shear strain curve and calculated by Eq. (4) for plain strain conditions (Bolton, 1986). The average of the dilatancy angles (under different confining stresses) calculated by Eq. (4) for each simulant is plotted in Fig. 14. Corresponding dilatancy angles are somewhat lower than expected values reported for dense specimens in literature based on direct shear and triaxial compression tests, which indicates that the target initial density selected for specimen preparation ($\rho = 1.68$ g/cm^3) more likely corresponds to medium dense state for the specimens in this study. At this point one can wonder why relative density (D_r) was not included among the density index parameters considered in Table 4. This is because the relevant standards (ASTM D4253, D4254, 2005 Japanese Geotechnical Society, 2000) for determination

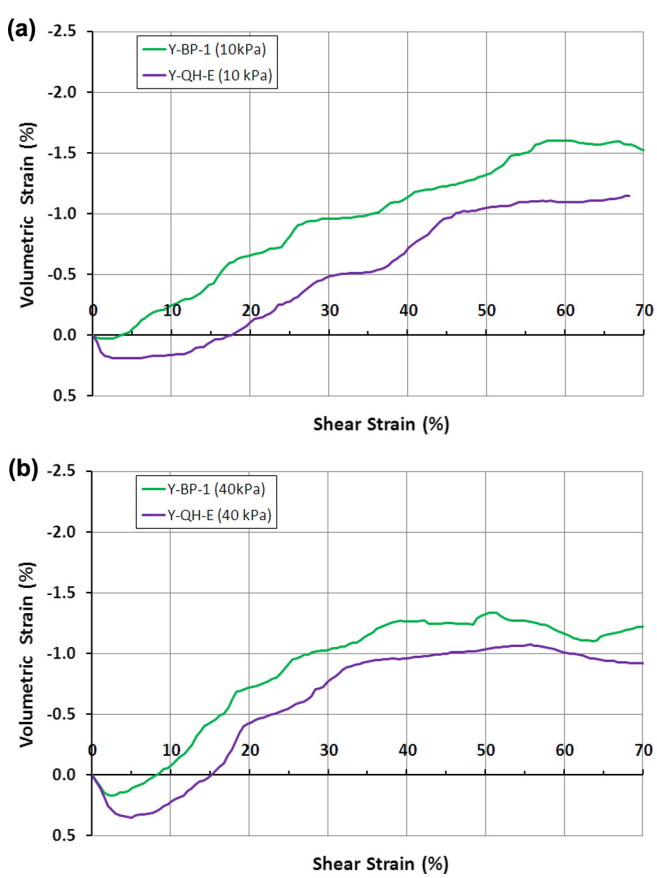

Fig. 13. Volumetric strain-shear strain responses of the first pair of lunar soil simulants: Y-BP-1 and Y-QH-E (a) under $\sigma_n = 10$ kPa, (b) under $\sigma_n = 40$ kPa.

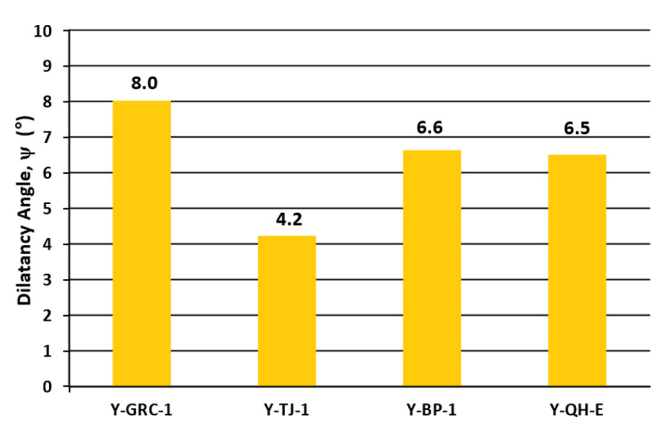

Fig. 14. Average dilatancy angles for the tested lunar soil simulants.

of e_{max} and e_{min} are only valid for soils with considerably lower fines contents than the vast majority of the lunar soil simulants (e.g. for ASTM procedures FC $\leq$ 15%, for Japanese Standards FC $\leq$ 5%). Indeed, this is probably the reason why bulk density (ρ) had been traditionally used in the previous studies (Table 3). Nevertheless, virtually all specimens have shown volumetric dilation (Figs. 12 and 13) and interesting observations could be made from Fig. 14. It could be seen for the first pair of simulants that ψ for Y-GRC-1 was almost double of the Y-TJ-1, whereas ψ for second pair of simulants were identical (Y-BP-1 and Y-QH-E).

$$\psi = \tan^{-1}\left(-\frac{d\varepsilon_v}{d\gamma}\right) \quad (4)$$

More explicitly, when the simulants had a sand dominated fabric (i.e. FC $\leq$ 25%) and had the same D_{50} (Y-GRC-1 & Y-TJ-1), increasing FC significantly reduces the rate of dilation when sheared at the same density. On the other hand, when the simulants had a silt dominated fabric (or transition towards a silt dominated fabric) (i.e. FC $\geq$ 40%) and had the same CU (Y-BP-1 & Y-QH-E), increasing FC had negligible effect on the rate of dilation when sheared at the same density. It should be reminded that the mentioned observation based on ψ express the rate of dilation, while the absolute magnitude of dilation represented by ε_v is still influenced by the amount of FC as explained before (see Fig. 13).

5. Consideration of limitations, practical implications of findings for planetary exploration and future research

Terrestrial soils are formed by physical and chemical weathering of various parent rocks and their fragments. Mitchell (1993) mentioned unloading, thermal expansion and contraction, crystal growth, colloid plucking and organic activity among the important processes of physical weathering. Meanwhile, weathering of lunar rocks has different processes such as impacts of macro and micro meteoroids, and bombardment by charged atomic particles from the Sun (Duke, 2006; Gaier, 2008). These "space weathering" processes result in the formation of extraterrestrial components in the lunar soil such as agglutinates. From geochemistry point of view, agglutinates are classified as mineral and glass combination, which contains minor amounts of metallic iron (Fe^0) and troilite (FeS). Detailed information about the element chemistry of the lunar parent rocks and mineral abundances is available in literature (Duke, 2006; Sibille et al., 2006). From geotechnical point of view, agglutinates could draw attention because they are weaker than many other mineral grains (Gaier, 2008), hence could influence the frictional response of the lunar soil.

XRF (X-ray fluorescence) analysis was used to determine the elemental composition of the tested proxy simulants. Table 5 gives a general comparison of the chemical composition of the sand, silt constituents of the proxy simulants tested in this study with the average Apollo 14 soil and JSC-1 lunar soil simulant. As seen in Table 5, mineral compositions of the proxy simulants tested in this study have differences when compared with JSC-1 and Apollo 14 soils. More specifically, proxy simulants have considerably greater quartz (SiO_2) content, while some other minerals are either absent (e.g. FeO) or considerably smaller (e.g. MgO), as expected for many terrestrial sandy soils. Authors are well aware of the fact that bulk chemistry and mineralogy of a material is quite important for many reasons, however it should be reminded one more time that the goal of the present study is to investigate the potential impacts of the grain size distribution only, by isolating the effects of several other factors including the mineralogy and morphology. In fact, the problem is quite complex and perhaps this (mixing a base sand with silt to form proxy simulants) is a simple, yet better approach to focus on the impacts of the grain size distribution only, considering the following aspects:

Table 5
The elemental composition of the tested proxy simulants from XRF (X-ray fluorescence) analysis.

Oxide	Sile Sand (% weight)	IZ Silt (% weight)	JSC-1[a] (% weight)	Average[a] Apollo 14 soil
SiO_2	97.207	68.691	47.71	48.1
TiO_2	0.163	0.706	1.59	1.7
Al_2O_3	1.242	11.438	15.02	17.4
FeO	–	–	7.35	10.4
Fe_2O_3	0.738	–	3.44	–
MgO	0.036	1.5	9.01	9.4
CaO	0.075	10.364	10.42	10.7
Na_2O	0.055	0.992	2.70	0.7
K_2O	–	1.756	0.82	0.55
MnO	0.018	0.134	0.18	0.14
Cr_2O_3	0.197	0.077	0.04	0.23
P_2O_5	0.012	0.143	0.66	0.51

[a] Data from Sibille et al. (2006).

(1) Many of the simulants summarized in Table 1, including the tested pairs (Table 2), already have variations in mineralogy when compared with each other and/or Apollo soils as well. To give few examples, GRC-1 was produced by mixing four different terrestrial sand blends obtained from a local sand company at Ohio, USA (Oravec et al., 2010). Whereas, TJ-1 was produced from the red volcanic ash deposits at Jilin Province, China (Jiang et al., 2012). On the other hand, BP-1 was obtained from the silty tailings remained from aggregate production mainly for highway projects at Arizona, USA, which was reported to behave very much like lunar soil (Rahmatian and Metzger, 2010). Therefore, it is not surprising to expect that mentioned simulants would have mineralogical differences. Recently, Taylor et al. (2016) mentioned about the disparity between chemical compositions of various popular lunar soil simulants and had interesting discussions on some of the

exaggerations about their properties. As an example both JSC-1 and JSC-1A, known as good mare soil simulants by lunar research community (also commonly investigated from geotechnical perspective as seen in Table 3), were stated by Taylor et al. (2016) as not properly resembling the chemical composition of the mare soils. When agglutinates are of concern, some simulants (e.g. GRC-1, MLS-1) did not contain any glassy agglutinate fraction (Sibille et al., 2006) such as the proxy simulants tested in this study.

(2) Another aspect is the differences in morphology of the different lunar soil simulants, which could also influence the resulting stress strain response. As an example, Rahmatian and Metzger (2010) reported that the mechanical behavior of JSC-1A (perhaps the most commonly investigated simulant for geotechnical characterization, see Table 3) is different than other lunar soil simulants because of having more rounded grain shape.

(3) The third aspect is the relativity of fidelity for different applications (e.g. geotechnical, toxicity, mining, abrasion, radiation shielding etc.) as also mentioned by many experts (Duke, 2006; Gaier, 2008). Taylor et al. (2016) wrote about an interesting example about geotechnical usage. Accordingly, a mixture of a sand and kaolinite was used as a lunar soil simulant during the Apollo time to conduct research on drilling into lunar soil at Johnson Space Center. With today's scientific knowledge about lunar soils, it is hard to call it a lunar soil simulant, but it did the job for the specific application (Taylor et al., 2016).

Nevertheless, proper consideration of some limitations explained above imply that the absolute magnitudes of several geotechnical parameters (e.g. c, ϕ, ψ) of the proxy simulants tested in this study could change, had a better mineralogical and/or morphological match with the lunar soil was obtained. However, trends observed and discussed in this study will most likely remain to be valid. In other words, experimental results in this study clearly demonstrated that gradational characteristics of the lunar soil simulants influence their shear strength and volume change behaviors. Therefore, such effects should be carefully evaluated when the future simulants will be produced and utilized for experimenting several engineering purposes including lunar exploration (landing, terramechanics of rovers) and lunar construction activities (e.g. site preparation, lunar bases, transportation networks etc.). Similarly, it is not surprising to expect that the findings in this paper could also be useful in general terms regarding the stress-strain behavior of Martian soil simulants with the assumption that similar soil mechanics principles would be valid, although conditions at the Moon and Mars are not necessarily the same.

In order to confirm and fine tune the findings in this study, space agencies and research groups are suggested to test lunar soil simulants adjusted to different grain size distributions with controlled gradation parameters (e.g. D_{50}, CU, FC etc.), but this time matching the bulk chemistry and grain shape characteristics with the interested Apollo samples. That would also help to compare how terrestrial and extraterrestrial fines content would influence the micro fabric and the resulting stress-strain behavior.

6. Summary and conclusions

In this study the influence of grain size distribution on the stress-strain behavior of lunar soil simulants were investigated. In the first part of the study, grain size distribution characteristics of the vast majority of the lunar soil simulants produced by different research groups from various countries were compiled and compared (Fig. 1, Table 1). Two pairs of simulants (four simulants) were selected based on the analyzed gradation characteristics and the selected simulants were reproduced in terms of very similar grain size distribution characteristics by mixing a terrestrial sand and silt. The first pair of reconstituted proxy simulants (Y-GRC-1 and Y-TJ-1) had almost identical mean grain size ($D_{50} \approx 0.28$ mm) but considerably different CU (i.e. CU of Y-TJ-1 is six times that of Y-GRC-1), while the second pair (Y-BP-1 and Y-QH-E) had identical CU ($\approx$13.5) but different D_{50}. Simulants prepared at the same initial density (i.e. $\rho = 1.68$ g/cm^3) were then tested in simple shear conditions first time in literature.

Drained shear strength characteristics of the tested proxy simulants were compared based on the Mohr-Coulomb failure criterion at the peak and residual states. When the first pair of simulants is considered; drained shear strength and peak secant friction angles ($\phi_{\text{sec-peak}}$) of Y-GRC-1 was found to be slightly greater than those of Y-TJ-1. This is an interesting observation because with many major factors (e.g. density, D_{50}, initial stress conditions, mineralogy, grain shape etc.) being the same, one would normally expect a very well graded soil such as Y-TJ-1 (with CU = 25) to have greater shear strength than a poorly graded soil such as Y-GRC-11 (with CU = 4.1). When the second pair of the proxy simulants (with identical CU) is considered; shear strength and $\phi_{\text{sec-peak}}$ were observed to increase with decreasing D_{50}.

When volume change behaviors of the first pair of the proxy simulants were compared, there was a clear trend of Y-GRC-1 specimens being considerably more dilative than the T-TJ-1 specimens both at the same density and at the same void ratio. This is also a controversial observation because with many major factors (e.g. density, D_{50}, initial stress conditions, mineralogy, grain shape etc.) being the same, one may expect a very well graded soil such as Y-TJ-1 (with CU = 25) to be more dilative, due to the possibility of more efficient packing tendency and interlocking effect between the grains, than a poorly graded soil such as Y-GRC-11 (with CU = 4.1). When the second pair of the proxy simulants (with identical CU) is considered; Y-BP-1 specimens (with greater D_{50}) were observed to be

more dilative than the Y-QH-E specimens (with smaller D_{50}).

It was shown that possible reasons for the above mentioned observations (including the controversial ones) are highly related with the initial fabric of the simulant specimens before shearing. The proportion and positional tendencies of the silt grains and sand grains within the simulants were explained to influence the initial fabric and therefore the stress-strain behavior of the proxy simulants significantly. As an example, even though the amount of volumetric dilation had decreased with increasing fines content for both simulant pairs, it was shown that the first pair of simulants (Y-GRC-1 and Y-TJ-1) had a sand dominated fabric, while the second pair of simulants (Y-BP-1 and Y-QH-E) had a silt dominated (or transition towards a silt dominated) fabric.

In the last part of the study, the rate of volumetric dilation of simulants was investigated via the magnitude of the dilatancy angle (ψ). It was determined that when the simulants had a sand dominated fabric, had the same D_{50} and density, increasing FC significantly reduced the rate of dilation. On the other hand, when the simulants had a silt dominated fabric (or transition towards a silt dominated fabric), had the same CU and density, increasing FC had negligible effect on the rate of dilation.

The findings and conclusions in this study show that gradational characteristics of the lunar soil simulants influence their shear strength and volume change behaviors. Therefore, such effects should be carefully evaluated when simulants are produced and utilized for experimenting several engineering purposes including lunar exploration (landing, terramechanics of rovers) and lunar construction activities (e.g. site preparation, lunar bases, transportation networks etc.). Lastly, the findings in this paper could also be useful in general terms regarding the stress-strain behavior of Martian soil simulants with the assumption that similar soil mechanics principles would be valid, although conditions at the Moon and Mars are not necessarily the same.

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